

Literature Status Note for arXiv:math/0404304

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Status. `literature_already_answered` (non-original; the source itself footnotes the announced positive answer, and a later paper gives a theorem-level write-up).

Source question. In Section 4, “Open Problems”, item (b), Brudnyi and Brudnyi ask whether the conclusion

$$\lambda(\ell_p) = \infty \quad (p \neq 2)$$

also holds for infinite-dimensional Hilbert spaces. The same item contains a footnote saying that the answer is positive and that a forthcoming paper will prove

$$\lambda(X) = \infty$$

for every infinite-dimensional Banach space X .

Supporting paper and theorem. Sofi, *Extension operators and nonlinear structure of Banach spaces*, arXiv:2001.09303, gives the relevant statement in Theorem 2.10: a Banach space X has the simultaneous Lipschitz extension property (SLEP) if and only if X is finite dimensional. In the proof, Sofi explicitly claims and derives that

$$\lambda(X) = \infty$$

for every infinite-dimensional Banach space X , using Dvoretzky’s theorem, lower bounds for $\lambda(\ell_2^n)$, and the retraction inequality for extension constants. Therefore the source question has a positive answer for $X = \ell_2$, and in fact for every infinite-dimensional Hilbert space.

Why this is not a new result. The source question is already accompanied by a footnote announcing the positive answer. The supporting paper supplies a later explicit theorem-level account and cites the Brudnyi–Brudnyi literature for the underlying nonextension argument. This packet is therefore only a duplicate-prevention and literature-status record.

Scope notes. The theorem also shows that any infinite-dimensional Hilbert space fails SLEP. Consequently, if the source’s later Alexandrov open problem is read without a finite-dimensional or local-compactness restriction, then $M = \ell_2$ gives a complete simply connected CAT(0) length-space obstruction to $\lambda(M) < \infty$. This packet does not address a possible narrower interpretation restricted to finite-dimensional or locally compact Alexandrov spaces. It also does not settle the source’s general product problem or uniform-lattice conjecture.

Local evidence. The source question is visible in the parsed TeX for arXiv:math/0404304 around lines 1072–1083. Sofi’s SLEP theorem and proof are visible in the parsed TeX for arXiv:2001.09303 around lines 231–283.